



ArenaMind AI — 2-Hour MVP Development Roadmap

Hackathon Strategy

Goal: Build a visually stunning, AI-powered prototype that demonstrates the product vision—not every production feature.

Target: Deliver a polished MVP capable of impressing judges within **2 hours**.



MVP Success Criteria

By the end of 2 hours, the prototype should demonstrate:

- ☒ Beautiful Landing Page
- ☒ AI Command Center
- ☒ Interactive Digital Twin (Basic)
- ☒ AI Chat (Gemini + Groq Fallback)
- ☒ Crowd Prediction Demo
- ☒ Emergency Response Demo
- ☒ Smooth Navigation
- ☒ Premium Animations



Timeline

0:00 —————> 2:00



Phase 1 (0:00 – 0:20)

Project Foundation (20 Minutes)

Setup

- Create Next.js Project
 - Install Tailwind CSS
 - Install Shadcn/UI
 - Install Framer Motion
 - Install React Three Fiber
 - Install Drei
 - Install Lucide Icons
 - Install Zustand
 - Configure Environment Variables
-

Folder Structure

```
app/  
components/  
lib/  
hooks/  
services/  
store/  
public/  
styles/
```

Deliverables

- ✓ Project running
- ✓ Theme configured

✓ Global Layout

Phase 2 (0:20 – 0:50)

Build Landing Experience (30 Minutes)

Create

- Landing Page
 - Hero Section
 - Floating Navigation
 - AI Branding
 - CTA Buttons
 - Feature Cards
 - Footer
-

Add

- Framer Motion
 - Glassmorphism
 - Gradients
 - Smooth Scrolling
-

Deliverables

✓ Landing Page

Phase 3 (0:50 – 1:20)

AI Command Center (30 Minutes)

Build

- Dashboard Layout
- Sidebar
- KPI Cards
- AI Summary
- AI Chat
- Notification Panel

Mock Data

Attendance

Crowd

Transport

Incidents

Energy

Deliverables

✓ Command Center

Phase 4 (1:20 – 1:40)

AI Integration (20 Minutes)

Implement

Gemini

↓

Fallback

↓

Groq

Example Prompt

```
Predict crowd congestion  
near Gate B.
```

Return

```
Prediction  
  
High  
  
Confidence  
  
96%  
  
Recommendation  
  
Open Gate D.
```

Deliverables

- ✓ AI Chat
- ✓ Streaming Response
- ✓ Groq Failover

Phase 5 (1:40 – 2:00)

Digital Twin + Demo Polish (20 Minutes)

Build

- Stadium Model (basic)
- Clickable Hotspots
- Crowd Overlay

- Emergency Highlight
- AI Recommendation Panel

Polish

- Animations
- Hover Effects
- Loading States
- Responsive Layout

Deliverables

- ✓ Interactive Digital Twin
- ✓ Demo Ready

MVP Features

Feature	Status
Landing Page	✓
AI Command Center	✓
AI Chat	✓
Gemini Integration	✓
Groq Fallback	✓
Digital Twin	✓
Crowd Prediction	✓
Emergency Demo	✓
Responsive UI	✓

Skip During Hackathon

Do NOT build:

- Authentication
- Firestore Database
- WebSockets
- Real CCTV Integration
- RAG
- Multi-Agent System
- Real Analytics
- Live Sensors
- Notifications Backend
- Reports Export
- Multi-language Engine
- Real Maps Routing

Use mocked/demo data instead.

Tech Stack

Frontend

- Next.js
- Tailwind CSS
- Shadcn/UI
- Framer Motion
- React Three Fiber

AI

- Gemini 2.5
- Groq (Fallback)

3D

- Three.js
- Drei

Deployment

- Vercel
-

Demo Flow (2–3 Minutes)

1. Open Landing Page
 2. Enter AI Command Center
 3. Show Digital Twin
 4. Ask AI:
 - "Predict crowd congestion near Gate B."
 5. Display AI recommendation
 6. Trigger Emergency Demo
 7. Show Crowd Heatmap
 8. End with Executive Dashboard preview
-

Final Deliverable

By the end of 2 hours, you should have a polished prototype that demonstrates:

- AI-powered operational intelligence
- Interactive Digital Twin
- Crowd prediction
- Emergency response
- Modern UI/UX
- Gemini + Groq failover

Focus on **experience over completeness**. A seamless, visually impressive demo with one or two working AI workflows will leave a much stronger impression than an unfinished implementation of every planned feature.

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Build Less. Demo Better. Impress More.

</div>